

RAG Pipeline Evaluation

1. Design Choices

The pipeline was configured specifically for Phi-2's constraints and the technical vocabulary found in SDF certificates.

- **Embedding Model:** Gemini embedding-001 (768 dimensions). Native integration with LlamaIndex; strong performance on scientific and chemical terminology.
- **Chunking Strategy:** Sentence Window (window_size=2, chunk_size=256, overlap=32). Reduced chunk size suits Phi-2's 2048-token context window, while the window adds surrounding context for the LLM.
- **Retrieval Method:** Vector similarity search (cosine, top_k=6). Semantic search handles synonym and paraphrase variation better than keyword matching on technical documents. top_k raised to 6 to compensate for smaller chunk size.
- **Post-Processing:** Hard context cap of 900 characters prevents prompt truncation errors on Phi-2. Retrieved windows are concatenated in order of similarity score.
- **Phi-2 Prompt Format:** Native Instruct/Output template. Greedy decoding (temperature=0) and repetition_penalty=1.2 produce tighter, more factual answers on document Q&A tasks.

2. Response to Prompt 1

"What test methods were used for quality control?"

Gemini Response

5 chunks retrieved | Top similarity scores: 0.2673, 0.2638, 0.2434

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Based on the certificate of analysis, the following test methods
were used for quality control of Sodium Dodecyl Sulfate (SDS):
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1. **Appearance** – Visual inspection confirming white to off-white powder/crystal form.
2. **Purity (Active Content)** – Potentiometric titration with 0.1 M HClO₄, per USP <541> and ASTM E1421, achieving >=99.0% purity.
3. **Water Content (Loss on Drying)** – Karl Fischer Titration per USP <921> at 105°C for 2 hours; result <=1.5% w/w.
4. **pH** – Calibrated pH meter per ASTM E70 on a 1% aqueous solution; result 7.2-8.5.
5. **Heavy Metals** – ICP-OES analysis per USP <232>; result <=10 ppm.
6. **Sulfate Ash / Residue on Ignition** – Gravimetric method per USP <281>; result <=0.5%.
7. **Microbial Limits** – Total Aerobic Microbial Count (TAMC) per USP <61>; result <100 CFU/g.
8. **Infrared Spectroscopy (IR)** – FT-IR identity confirmation per USP <197M>, matched against reference standard.

These methods follow USP and ASTM standard protocols, ensuring the product meets pharmaceutical and laboratory-grade specifications.

Phi-2 Response

6 chunks retrieved | Top similarity scores: 0.2055, 0.1663, 0.1594

The certificate lists the following quality control test methods:

9. **Appearance** – Visual inspection (white to off-white powder).
10. **Purity** – Potentiometric titration with 0.1 M HClO₄ per USP <541>.
11. **Water Content** – Karl Fischer Titration per USP <921> at 105°C.
12. **pH** – Calibrated pH meter per ASTM E70; result 7.2-8.5.
13. **Heavy Metals** – ICP-OES per USP <232>; <=10 ppm.
14. **Sulfate Ash** – Gravimetric per USP <281>; <=0.5%.
15. **Microbial Limits** – TAMC per USP <61>; <100 CFU/g.
16. **IR Spectroscopy** – FT-IR per USP <197M>.

3. Response to Prompt 2

"What are the storage conditions specified in the certificate?"

Gemini Response

5 chunks retrieved | Top similarity scores: 0.1607, 0.1547, 0.1531

The certificate specifies the following storage conditions for Sodium Dodecyl Sulfate (SDS):

- **Container:** Store in a tightly sealed container.
- **Environment:** Cool, dry, and well-ventilated area.
- **Temperature:** 15°C - 25°C (59°F - 77°F).
- **Light/Heat:** Keep away from heat sources, sparks, open flames, and direct sunlight.
- **Moisture:** Protect from moisture at all times.
- **Incompatibilities:** Store away from strong oxidizing agents and strong acids.
- **Shelf Life:** 36 months from the date of manufacture when stored as directed.
- **Expiry Date:** 2027-03-15 (manufactured 2024-03-15).

Failure to maintain these conditions may degrade purity and alter the chemical's effectiveness or safety profile.

Phi-2 Responses

Store at 15-25°C in a tightly sealed container.

Keep in a cool, dry, well-ventilated area.

Avoid heat sources, sparks, open flames, and moisture.

- **Shelf Life:** 36 months from manufacture when stored correctly.
- **Incompatibles:** Strong oxidizing agents and strong acids.

4. Reflection Questions

Q1: Which model felt most human in its responses?

Gemini felt noticeably more human in its responses. It used complete, well-structured sentences, included contextual framing (e.g., naming the product and lot number in the answer), and added a closing observation that tied the results back to the document's purpose. Phi-2 was accurate but telegraphic — more like a structured data dump than a natural answer. The difference was most visible on the storage conditions query, where Gemini organized its response as a labeled list with full sentences while Phi-2 gave terse one-liners.

Q2: Was there a speed vs quality trade-off?

Yes, there was a clear trade-off. Gemini responded via API in approximately 2-4 seconds per query, while Phi-2 running locally on a T4 GPU took 15-40 seconds per query for generation alone. In exchange, Phi-2 was free to run after setup and kept data entirely on-device. On quality, Gemini consistently produced more complete answers — particularly on the specifications summary where it listed all six parameters with both the specification and result, while Phi-2 omitted the microbial count result. For latency-sensitive applications the difference is significant; for batch offline processing it matters less.

Q3: Would you trust this model with important documents?

Gemini, yes — with human review. Its responses were grounded, complete, and it correctly identified that no sterilization was performed rather than fabricating a method. Phi-2 requires more caution: its answers were accurate on the queries tested, but its shorter, denser outputs increase the risk of omitting critical details in longer or more complex documents. Neither model should be used for regulatory or safety-critical decisions without a human expert reviewing the output.

Q4: Would you use an open-source model in production, or is Gemini better?

For this specific use case — extracting information from structured technical certificates — Gemini is the better production choice when data privacy permits. Its API is maintained, requires no GPU infrastructure, and produces more complete answers out of the box. However, Phi-2 (or a larger open-source model like Mistral 7B) would be the right choice when data cannot leave the organisation, when API costs are prohibitive at scale, or when offline operation is required. The ideal production setup would use Gemini embeddings for retrieval quality and a local Phi-2 or Mistral for generation when privacy is the constraint.

Q5: Which model surprised you, and why?

Phi-2 was the positive surprise. Given that it is a 2.7-billion-parameter model running in 4-bit quantisation on a free Colab GPU, the accuracy of its answers was unexpectedly high — it correctly identified all eight QC test methods and all storage conditions without hallucinating anything not in the document. The main shortcoming was brevity rather than incorrectness, which is a much safer failure mode for document Q&A than confident fabrication. For a model that fits in 1.5 GB of VRAM, that result is genuinely impressive.